BROAD INSTITUTE

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To Whom It May Concern:

I am writing to express my strongest possible support for the application of Matthieu Broisin for the doctoral program. He brings together an incredible combination of skills in engineering and mathematics that make him ideally positioned to become a strong biomedical researcher.

For context, my laboratory at the Broad Institute of Harvard and MIT develops advanced methods and software tools to quantify and mine the rich information present in cellular images to yield biological discoveries. Our laboratory is best known for our freely available software packages CellProfiler and CellProfiler Analyst, which are continuously improved based on our collaborations with dozens of researchers around the world and have been cited in more than 7,000 papers, and used extensively in academia and pharma. My laboratory is also pioneering image-based profiling, where thousands of morphological features are measured from each cell's image to create a profile, or signature, of cell state for comparison to other cell populations of interest (e.g. across drug treatments or disease states).

Through my experiences, I have been fortunate to work with a number of brilliant and hard-working individuals. Matthieu is, without question, in the top tier I have worked with. I say this for several reasons.

First, having spent my professional career in the field of biological imaging, I am very familiar with the quality of his work. It is rare that someone of Matthieu's engineering and mathematical caliber enters into the life sciences, and I am confident that he will make deep and significant contributions. He chose Allen Goodman in my group as a mentor. Allen was able to articulate deep and open questions in computer vision and machine learning to Matthieu to get him up to speed quickly.

Second, I am very impressed by Matthieu's commitment to biomedical research, as I learned through our career discussions as he was deciding whether to switch fields. It was not easy to change fields. It takes courage and dedication. In the case of Matthieu, his desire comes from a deep desire for his work to have bigger impact on the human condition. As he learned about the field of biological imaging, he realized that there are problems to solve that are both intellectually rich and can also have a significant impact on our understanding of disease. It is this desire for greater impact, and a genuine commitment to biomedical research, that drives his current career choices. The next phase of biomedical research will require an influx of researchers from computational domains, and we are lucky that an intellect of Matthieu's caliber will be one of them.

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Finally, at a personal level, Matthieu possesses a collaborative spirit, a humble disposition, a sincere desire to learn new domains, and the intellectual honesty to ask questions when he doesn't understand. These are the human traits that make for an outstanding researcher, especially in an interdisciplinary field like biological imaging. He had no conflicts in the lab and worked well individually and as a team. I give him my absolute highest recommendation.

I hope you will look favorably on Matthieu's application. I know he will benefit tremendously from the doctoral program. He is new to the domain of biology and, while I am confident that he has the capacity to become a tremendous researcher, he will benefit greatly from the opportunity for further training in biology and image processing to complement his engineering and mathematical background. The doctoral program is ideally suited to someone of Matthieu's background.

Please do not hesitate to reach out to me with any additional questions.

Sincerely,



Anne Carpenter, Ph.D.
Institute Scientist and Imaging Platform Director, Broad Institute of MIT and Harvard
Co-Founder and leader, CellProfiler project
Merkin Institute Fellow

and

Allen Goodman
Senior Software Engineer, Broad Institute of MIT and Harvard